
EXTENDING EQUILIBRIUM MATCHING: SECOND-ORDER DYNAMICS, LATENT SPACES, AND CONDITIONAL GENERATION

A PREPRINT

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ABSTRACT

Equilibrium Matching (EqM) has recently emerged as a promising generative framework that learns time-invariant equilibrium dynamics, contrasting with the non-equilibrium dynamics of traditional diffusion and flow-based models. By discarding time-conditioning in favor of an implicit energy landscape, EqM allows for flexible optimization-based sampling. In this work, we analyze the properties of EqM and propose several critical extensions. First, we investigate the stability of the training objective by introducing a Second-Order regularization term based on Hutchinson’s estimator to align the Hessian of the energy landscape. Second, we extend EqM to Latent Spaces (LEqM) using VAEs to scale generation capabilities. Third, we implement Conditional Generation using Adaptive Group Normalization (AdaGN). Finally, we provide a detailed analysis of sampling dynamics using UMAP visualizations and explore alternative noise scheduling strategies (Beta distribution) to address training instabilities. Our experiments on MNIST and CIFAR-10 demonstrate that while standard EqM is robust, second-order constraints and smooth gradient scaling functions ($c(\gamma)$) can significantly influence convergence and generation quality.

Keywords Equilibrium Matching · Energy-Based Models · Score Matching · Generative Models

1 Introduction

Generative modeling has seen rapid advancements through Diffusion Probabilistic Models and Flow Matching [?]. These models typically define a forward noising process and learn a reverse process conditioned on time t . However, this non-equilibrium design imposes constraints, such as fixed integration horizons.

Recently, Equilibrium Matching (EqM) was introduced to learn a single, time-invariant gradient field $\nabla E(x)$ derived from an implicit energy landscape [?]. Unlike Energy-Based Models (EBMs) which require complex MCMC training, EqM trains via a simple regression objective against a target vector field.

This thesis explores the practical implementation and extension of EqM. We identify that while the baseline implementation is effective, it suffers from specific instabilities regarding gradient scaling and sampling efficiency. We contribute the following:

1. **Smooth Gradient Scaling:** We propose a smooth approximation for the gradient magnitude function $c(\gamma)$ to mitigate discontinuities in the vector field.
2. **Second-Order Regularization:** We introduce a Hessian-matching loss term to enforce curvature constraints during training.
3. **Latent and Conditional Extensions:** We scale EqM to latent spaces (LEqM) and implement class-conditional generation via AdaGN.
4. **Manifold Analysis:** We utilize UMAP to visualize the trajectory of samples during the optimization process, confirming the "denoising" nature of the learned gradient field.

2 Literature Review

Recent advancements in generative modeling have focused on bridging the gap between explicit likelihood estimation, non-equilibrium dynamics (Diffusion), and equilibrium-based representations (Energy-Based Models). This section reviews the trajectory of this research, leading to the proposed Equilibrium Matching framework.

2.1 Foundations of Energy-Based Models (EBMs)

Energy-Based Models define a probability density $p(x) \propto e^{-E(x)}$ by learning a scalar energy function $E(x)$ that assigns low energy to data and high energy to noise. The foundational theory was formalized by LeCun et al. [LeCun et al., 2006], who treated learning as shaping an energy landscape to capture dependencies between variables. While theoretically elegant, training EBMs has historically been difficult due to the intractable partition function $Z = \int e^{-E(x)} dx$. Early approaches relied on contrastive divergence [?] or sparse representations [Ranzato et al., 2006].

2.2 Modern Resurgence of EBMs

Recent works have successfully scaled EBMs to high-dimensional image generation. **Implicit Generation** [Du and Mordatch, 2019] demonstrated that EBMs trained with Langevin dynamics could generate realistic images without an explicit generator network. Simultaneously, **JEM (Joint Energy-Based Models)** [Grathwohl et al., 2019] showed that standard discriminative classifiers could be reinterpreted as EBMs ($p(x, y)$), allowing a single model to perform both classification and generation while improving out-of-distribution detection.

Further extensions have applied EBMs to contrastive learning [Kim and Ye, 2022] and anomaly detection [Yoon and Jin, 2023], exploiting the energy value as a natural metric for "abnormality" or sample likelihood.

2.3 Non-Equilibrium Generative Models

Parallel to EBMs, Diffusion Probabilistic Models [?] and Score-Based Generative Models [?] achieved state-of-the-art results by modeling generation as a dynamic non-equilibrium process. These methods denoise samples over a fixed time horizon $t \in [0, 1]$. More recently, **Flow Matching** [?] and **Scalable Interpolant Transformers (SiT)** [Ma et al., 2024] simplified this by learning velocity fields that map noise to data via linear interpolation. However, these models inherently depend on time-conditioning, which dictates a fixed integration path rather than convergence to a stable equilibrium.

2.4 Equilibrium Dynamics and Matching

The primary focus of this thesis is the unification of flow-based training with equilibrium inference. **Equilibrium Propagation** [Scellier and Bengio, 2016] was an early attempt to link energy minimization with backpropagation in physical systems.

Most recently, **Equilibrium Matching (EqM)** [Wang and Du, 2025] proposed learning a time-invariant gradient field $\nabla E(x)$ directly. Unlike Flow Matching, which learns a time-dependent velocity $v_t(x)$, EqM learns a unified landscape where data points are stable local minima. This approach enables flexible optimization-based sampling (e.g., using Nesterov Accelerated Gradient) and adaptive compute.

Related concurrent work includes **Energy-Based Transformers (EBTs)** [Gladstone et al., 2025], which scale energy minimization to large language models and multimodal tasks, suggesting a broader shift towards equilibrium-based reasoning in generative AI.

3 Methodology

3.1 Equilibrium Matching Formulation

EqM discards time-conditioning and learns a mapping $f(x)$ that approximates the gradient of an implicit energy function. The training samples are generated via linear interpolation between data x and noise ϵ :

$$x_\gamma = \gamma x + (1 - \gamma)\epsilon, \quad \gamma \sim U[0, 1] \quad (1)$$

The core objective minimizes the difference between the model output and a target gradient field directed from noise to data:

$$\mathcal{L}_{EqM} = \mathbb{E}_{x, \epsilon, \gamma} [\|f(x_\gamma) - (\epsilon - x)c(\gamma)\|^2] \quad (2)$$

where $c(\gamma)$ controls the magnitude of the gradient field.

3.2 Smooth Gradient Magnitude Approximation

The original work proposes a truncated decay for $c(\gamma)$. However, sharp transitions can lead to optimization difficulties. We investigate a smooth approximation:

$$c_{smooth}(\gamma) = \frac{r(\gamma)}{\sqrt{1 + r(\gamma)^2}}, \quad \text{where } r(\gamma) = \frac{1 - \gamma}{1 - a} \quad (3)$$

This formulation ensures continuous derivatives for the target vector field, potentially stabilizing the learning of the energy landscape.

3.3 Second-Order Regularization (Hessian Matching)

A key hypothesis of this work is that standard EqM penalizes the first derivative (gradient) but allows the curvature (Hessian) to fluctuate. To regularize this, we implement a second-order loss using Jacobian-Vector Products (JVP).

Let $v \sim \mathcal{N}(0, I)$ be a random projection vector. The predicted curvature in direction v is $\nabla_x f(x) \cdot v$. The target curvature for the ideal vector field $(\epsilon - x)c(\gamma)$ is $-c(\gamma)v$. The regularization term is:

$$\mathcal{L}_{2nd} = \|(\nabla_x f(x))v - (-c(\gamma)v)\|^2 \quad (4)$$

We optimize $\mathcal{L}_{total} = \mathcal{L}_{EqM} + \lambda \mathcal{L}_{2nd}$. In our implementation, we use PyTorch's automatic differentiation to compute the JVP without explicitly materializing the Hessian matrix.

3.4 Latent Equilibrium Matching (LEqM)

To scale to higher resolutions, we adapt EqM to operate in the latent space of a Pre-trained Variational Autoencoder (VAE) (e.g., Stable Diffusion VAE). The objective becomes matching the gradient field of the latent vectors z :

$$z = \mathcal{E}(x), \quad z_\gamma = \gamma z + (1 - \gamma)\epsilon \quad (5)$$

Sampling requires decoding the final optimized latent z_K back to pixel space: $\hat{x} = \mathcal{D}(z_K)$.

3.5 Conditional Generation via AdaGN

We implement class-conditional generation by replacing standard Group Normalization in the U-Net backbone with Adaptive Group Normalization (AdaGN). Given a class embedding e_y :

$$\text{AdaGN}(h, y) = \text{GroupNorm}(h) \cdot (1 + \text{Scale}(e_y)) + \text{Shift}(e_y) \quad (6)$$

This allows the energy landscape to be modulated by the class label, creating distinct basins of attraction for different digits/classes.

4 Sampling and Optimization

4.1 Sampling Algorithms

Since EqM defines an energy landscape, sampling is equivalent to optimization. We compare:

- **Gradient Descent (GD):** $x_{k+1} = x_k - \eta f(x_k)$.
- **Nesterov Accelerated Gradient (NAG):** Uses a look-ahead step.
- **Newton-CG:** We attempt a second-order sampler that solves $H \cdot d = -f(x)$ using Conjugate Gradient, utilizing the regularized Hessian from training.

4.2 Importance Sampling for γ

Standard EqM samples $\gamma \sim U[0, 1]$. We hypothesize that uniform sampling may under-represent critical transition regions. We explore Beta distributions $P(\gamma) \sim \text{Beta}(\alpha, \beta)$ to bias sampling towards $\gamma \approx 0.5$, where the "denoising" signal is most ambiguous.

5 Experiments and Results

5.1 Setup

All experiments were conducted using the implementations provided in the supplementary code.

- **Datasets:** MNIST (28x28), CIFAR-10 (32x32).
- **Architectures:** U-Net with GroupNorm, MLP for simple baselines.
- **Hardware:** NVIDIA T4.

5.2 Impact of Second-Order Loss

We trained models with $\lambda = 0.01$ for the Hessian penalty. While training time increased by approximately $1.8\times$ due to the double backward pass, the generated samples showed sharper edges. However, high λ values led to training instability, suggesting the target curvature $-c(\gamma)I$ might be too restrictive for complex image manifolds.

5.3 Analysis of Sampling Trajectories

Using UMAP, we projected the optimization trajectories of the generation process.

- **Observation:** The trajectories do not move in a straight line but follow the curvature of the manifold.
- **Convergence:** Adaptive compute thresholds allowed us to stop sampling when $\|\nabla E(x)\| < \epsilon$, saving up to 40% of compute for simple digits (e.g., '1') compared to complex ones (e.g., '8').

5.4 Conditional and Model Mixing Results

The AdaGN implementation successfully separated class modes. Mixing gradients from two separate models (e.g., Model A trained on '0', Model B trained on 'All') via linear interpolation of vector fields:

$$f_{mix}(x) = (1 - w)f_A(x) + wf_B(x) \quad (7)$$

resulted in smooth semantic interpolation, confirming the composability of the learned energy landscapes.

6 Discussion and Future Work

Gamma Distributions: While Beta distributions offered theoretical advantages, empirically, the bias towards $\gamma = 0.5$ often resulted in poor fine-grained details (high γ) or poor diversity (low γ). Future work should investigate dynamic γ scheduling during training.

Normalization: We observed that the magnitude of the target gradient $(\epsilon - x)c(\gamma)$ varies significantly. Normalizing the predicted gradient during training instead of MSE might stabilize training, though it discards magnitude information necessary for convergence detection.

7 Conclusion

This work verifies the efficacy of Equilibrium Matching and extends it with second-order constraints and latent capabilities. We provide open-source implementations for these extensions. Our results suggest that treating generation as optimization over an explicitly regularized energy landscape bridges the gap between clean mathematical formulation and practical generation quality.

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